Hooman Ramezani

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Experience

ML Specialist Solutions Architect Jan 2025 – Present
Nebius Group *Remote*

- Founding AI/ML specialist, helping neolabs move large-scale training and inference to the newest NVIDIA GPUs
- Lead pre-sales POCs on Blackwell and Vera Rubin for **$1B+** deals including assisting with software stack migration and performance optimization
- Inference optimization for Token Factory dedicated deployments with vLLM/SGLang on Blackwell and Vera Rubin

Machine Learning Researcher May 2023 – Dec 2024
University Health Network *Toronto, ON*

- First-author 'LN-Transformer' paper for lung nodule segmentation, achieving DiceC of 91.4% (SOTA)
- Built **multi-modal LLM** with unstructured text, CT slices for treatment recommendations with **84%** concordance

Machine Learning Engineer Sept. 2022 – April 2023
Advanced Micro Devices (AMD) *Toronto, ON*

- Cut **LLM inference** latency **80%** on AMD via quantization (QAT/PTQ), pruning, distillation, kernel tuning
- Benchmarked against **Triton** Inference Server (**TensorRT** backend) on k8s with dynamic batching

Machine Learning Engineer Jan. 2022 – April 2022
DarwinAI (Acquired by Apple) *Remote*

- Delivered medical vision model to Pfizer for advanced liver fibrosis diagnosis, achieving **91% sensitivity**
- Built end-to-end ML solutions for clients, with MLOps CI/CD pipelines for retraining and monitoring

Publications & Talks

Distributed KV Cache for AI Agents on NVIDIA B300 | *WEKA & Nebius Blogs* Sep 2026

- Led the Nebius side of bringing a distributed KV cache backend to market as a product on Nebius AI Cloud
- Benchmarked 512 concurrent agentic coding sessions: **2.4x** more requests on the same GPUs, **5.4x** faster first token

MXFP8 and DeepEP for 41% Faster DeepSeek-V3 Pre-training | *PyTorch Blog* Mar 2026

- Co-authored a joint blog post with **Meta's PyTorch team**, presented at NVIDIA GTC 2026
- Accelerated DeepSeek-V3 (671B) pre-training **41%** on 256 B200s with MXFP8 and DeepEP, no quality loss

LN-Transformer: Lung Nodule Transformer for Sparse CT Segmentation | *CVPR* Jun 2025

- Published a novel two-stage transformer for lung nodule segmentation featured in CVPR
- Strongest model on benchmark dataset **Dice 91.4%**, **F1 94.2%**, includes Meta SAM, DETR architectures

Education

University of Toronto Toronto, ON
MASc in Machine Learning & Operations Research *2022 – 2024*

- Thesis: Training small transformers for segmentation and detection; multimodal LLM and ViT for lung cancer

University of Waterloo Waterloo, ON
BASc in Systems Design Engineering, GPA: 3.7 / 4.0 *2017 – 2022*

Technical Skills

Languages: Python, C/C++, CUDA, Triton, SQL, Bash

Inference: vLLM, SGLang, TensorRT-LLM, NVIDIA Dynamo, LMCache, Triton Inference Server

Training: PyTorch, TorchTitan, Megatron-LM, FSDP, DeepEP, NVSHMEM, TorchAO (MXFP8), Hugging Face

Performance & Networking: Nsight Systems, Nsight Compute, NCCL, DCGM, InfiniBand / RDMA, NVLink

Infrastructure: Kubernetes, Slurm, Docker, Terraform, Linux, Git, AWS, GCP